

Dhairya P. Mishra

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AI/ML engineer with 4+ years building RLHF-adjacent evaluation pipelines, agentic systems, RAG infrastructure, and full-stack AI at scale. Co-author, Solaris (arXiv:2602.22208), NeurIPS 2026. Track record deploying customer-facing AI systems in production.

KEY SKILLS

- **AI/ML:** PyTorch, JAX, HuggingFace, RAG, RLHF, VectorDBs, Fine-Tuning, LoRA, Diffusion (DiT), LangChain, scikit-learn, FastAPI.
- **Languages:** Python, JavaScript/TypeScript, Node.js, React, Scala, Java, SQL, Bash, HTML/CSS.
- **Infrastructure & Cloud:** MLOps, AWS, GCP, Azure, GKE, Docker, Kubernetes, Apache Spark, ArgoCD, Terraform.
- **Data & Tools:** Kafka, Parquet, MongoDB, Git, Cursor, Claude, OpenTelemetry, Grafana, Playwright, Cassandra, Pydantic, Pandas.

EXPERIENCE

Evidenza Inc.

Jan 2026 – May 2026

AI Engineer (Intern) | Spark ETL, AlphaEvolve, FastAPI, multimodal scoring, Python, MCP, Prompt Engineering

Brooklyn, NY

- Productionized multi-step agent orchestration for LLM-driven persona generation using AlphaEvolve and FastAPI; achieved greater than 80% Monte Carlo coverage and 65% human-vs-synthetic agreement at scale..
- Built 6-stage Spark ETL pipeline via Spark; feeding LLM evaluation workflows powering a **new enterprise product capability**.
- Engineered multimodal evaluation pipeline across 1,200+ data samples; **quality-ranked boost of 18% inference accuracy**.

CVS Health

Jan 2023 – Jan 2025

Sr. Software Development Engineer | RAG, Hugging Face, ArgoCD, GCP, Terraform, Agile, OpenCV

New York, NY

- Built transformer-based inference pipeline for automated image content labeling using HuggingFace; downstream processing **defects reduced by 15% at scale**.
- Translated stakeholder requirements into an enterprise deployed **prompt-engineered RAG assistant** with Slack integration using GCP, OpenAI, and ChromaDB; **enabled self-service technical defect resolution** and **reduced open ticket time by 20%**.
- Reviewed and shipped **125+ PRs** and owned on-call with observability tooling, **reduced downtime by 12%** for customer-facing core platform systems (Digital-Blocks 2.0, Experience Builder) which served millions of customers daily.

Aetna Health

Feb 2022 – Jan 2023

Advanced Software Developer | Rally, Node.js, MongoDB, automation scripting

New York, NY

- Developed testing **automation suite** CAT, RallyScore, and ThemeScore for accessibility compliance; **reduced QA time by 75%**.
- Designed Rally kanban **migration pipeline** for 600+ nested structures; **reduced migration time by 80%**.

PUBLICATIONS

Solaris, Building a Multiplayer Video World Model in Minecraft – arXiv:2602.22208

Aug 2025 – Feb 2026

NeurIPS 2026 Submission | NYU VisionX, Diffusion Transformers (DiT), JAX, Flow Matching, Self Forcing, Multi-Agent Attention

- Designed **multi-agent evaluation framework** with **VLM as Judge** for movement consistency, spatial memory grounding, and cross-view coherence metrics; benchmarked generation quality across agent trajectories over **12.64M synchronized frames**.
- Introduced **Checkpointed Self Forcing** for **long-horizon training** at reduced memory cost; improved generation quality by 36% FID.

Testaro: Efficient Ensemble Testing for Web Accessibility – arXiv:2309.10167

Oct 2023

ACM SIGACCESS ASSETS 2023 | Node.js, Playwright, axe-core, WCAG/ADA compliance

- Represented CVS Health as **presenter for Testaro's** multi-tool accessibility evaluation to global research audience.

PROJECTS

Teserax (teserax.io) | React Flow, FastAPI, TypeScript, Zod, RAG, Playwright, EC2/Cloudflare

Mar 2026

- **Built from 0 to 1:** live LLM knowledge-graph app with **agentic multi-step orchestration**; AI auto-orchestrates **tool-based pipelines** over a DAG-safe React Flow canvas with **Brave Search Auto-RAG**; deployed on EC2 + Cloudflare, 226 Playwright E2E tests.

SliceWise, MRI Brain Tumor Detection & Segmentation | PyTorch, EfficientNet, U-Net, FastAPI, Streamlit, BraTS

Oct 2025

- Unified detection + segmentation into a single **shared-encoder model**; **91.3% acc, ~40% faster inference** vs two-model baseline.

PICO-LLM Research Pipeline (NYU) | PyTorch, KV-cache Transformer, LSTM, Gradio, wandb, top-p sampling

Apr 2025

- Designed multi-model **LLM hyper-param auto tuning pipeline** for small to medium datasets, achieved **73.21% token accuracy**.

EDUCATION

New York University – Courant

May 2026

M.S. Computer Science and AI | GPA: 3.63

New York, NY

Trine University

Dec 2021

B.S. Software Eng. and Mathematics

Angola, IN